PlayerArchitecture

1. Purpose

This document specifies the **player-specific** architecture built on the core character framework, focusing on:

- Input → Intent resolution (including multi-button modifiers)
- Intent arbitration (priority/consumption/buffering)
- Parallel domains (Traversal/Posture/CombatMode/Action) and their responsibilities
- Interaction pipeline (the "Circle button problem")
- Stealth + social actions (Reason/Bribe/Threaten) as first-class actions
- Cancel windows and buffering policy (high-level rules, not implementation)

This is intended to prevent state explosion and mixed concerns as scope expands (Dogma-style melee + Ratchet-like shooting + stealth + climbing/swimming + contextual interaction).

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2. Player Control Pipeline Overview

2.1 The control pipeline (conceptual)

1. **Raw Inputs** (engine bindings)
2. **InputSnapshot** (per-frame action states)
3. **IntentResolver** (context-independent mapping: actions/modifiers → candidate intents)
4. **InteractionResolver** (context-dependent mapping for "ContextInteract")
5. **IntentArbiter** (priority + consumption + buffering)
6. **Domains update** (Traversal, Posture, CombatMode, Action)
7. **Outputs + Events** (motor/anim/camera/sfx + gameplay events)

2.2 Why the pipeline exists

- Unity Input System answers: *what physical actions occurred?*
- The pipeline answers: *what should the character do, given context and constraints?*

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3. Core Concepts

3.1 InputSnapshot (raw action states)

The InputSnapshot contains **bound actions**, not physical device names:

- Movement: `Move`, `Look`
- Face actions: `Primary`, `Secondary`, `Tertiary`, `Jump`
- Modifiers: `SkillModifier` (R1), `AimModifier` (L1)
- Context: `ContextGrabOrFire` (R2), `ContextInteract` (Circle if not used as tertiary)

- Optional: `LockOn`, `SheatheToggle`, `Menu`, etc.

Each action is represented as `ButtonState`:

- `PressedThisFrame`
- `Held`
- `ReleasedThisFrame`

3.2 Candidate Intent

An intent is **game meaning**, not device meaning:

- `LightAttackIntent`
- `UseSkillIntent(slot, kind=Melee|Ranged)`
- `JumpIntent`, `DodgeIntent`
- `GrabOrClimbIntent`, `MantleIntent`
- `FireBasicRangedIntent`
- `ContextInteractIntent` (pre-resolution)
- `ReasonIntent(targetId)` (post-resolution)
- etc.

3.3 WorldSnapshot & InteractionContextSnapshot

- `WorldSnapshot`: locomotion/environment facts (grounded, water, climbable, mantle)
- `InteractionContextSnapshot`: best interaction candidates (door, NPC, loot), plus stealth/social opportunities.

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4. Parallel Domains

The player is always in multiple “states” concurrently. These are managed by **separate domains**.

4.1 Traversal Domain (TraversalFSM)

Responsibility

- Movement mode semantics:
  - `Grounded`, `Airborne`, `Swimming`, `Climbing`, `LedgeHang` (optional)
- Owns transitions driven by environment + movement intents:
  - Jump, land, enter water, grab ledge, start climb

Consumes intents

- `JumpIntent` (if applicable)
- `GrabOrClimbIntent` / `MantleIntent`
- Environment transition intents (if any)

Does not own

- Attacks/skills/social actions
- Equipment/aiming logic
- Quest interactions

4.2 Posture Domain (PostureFSM)

Responsibility

- Stance/collider profile:
  - `Standing`, `Crouched`, `Prone` (optional)
- Baseline stealth multipliers:
  - noise multiplier, visibility multiplier
- Movement caps related to posture (not full traversal)

Consumes intents

- `ToggleCrouchIntent` (or `PostureChangeIntent`)

Does not own

- Stealth takedown decision
- Interaction selection

4.3 Combat Mode Domain (CombatModeFSM)

Responsibility

- Weapon readiness and aiming mode:
  - `Sheathed`, `MeleeReady`, `RangedReady`, `Aiming` (or `IsAiming` flag)
- Controls which action set is active:
  - melee default attacks vs ranged default fire
  - skill slot type selection (melee vs ranged)

Consumes intents

- `AimStart/Stop` (or derives from `AimModifier`)
- `SheatheToggleIntent` (optional)
- `RequestUnsheatheIntent` (from ActionSystem)

Does not own

- Executing attacks/skills (ActionSystem owns)
- Traversal transitions

4.4 Action Domain (ActionSystem)

Responsibility

- Committed actions (attacks, dodge, skills, interact actions, social actions)
- Stamina costs, cooldowns, buffering, cancel windows
- Outputs movement/animation constraints while action runs

Consumes intents

- `LightAttackIntent`, `HeavyAttackIntent`
- `UseSkillIntent(...)`
- `DodgeIntent`
- `InteractIntent`, `LockpickIntent`, `PickpocketIntent`, `AssassinateIntent`, `KnockoutIntent`
- `ReasonIntent(targetId)` and other social intents

Does not own

- Physical grounded/swimming/climbing semantics
- Baseline posture selection

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5. Input → Intent Mapping (Dogma-inspired scheme)

5.1 Baseline (no modifiers)

- `Primary` → `LightAttackIntent`
- `Secondary` → `HeavyAttackIntent`
- `Tertiary` / `ContextInteract` → `ContextInteractIntent` (to be resolved)
- `Jump` → `JumpIntent`

5.2 SkillModifier held (R1 + melee weapon equipped)

- `Primary` → `UseSkillIntent(slot=1, kind=Melee)`
- `Secondary` → `UseSkillIntent(slot=2, kind=Melee)`
- `Tertiary` → `UseSkillIntent(slot=3, kind=Melee)`
- `Jump` → weapon-dependent:
  - dagger-like: `DodgeIntent`
  - otherwise: `JumpIntent`

5.3 AimModifier held (L1 held / bow aiming)

- `ContextGrabOrFire` (R2) → `FireBasicRangedIntent` (no stamina)
- `Primary/Secondary/Tertiary` → `UseSkillIntent(slot=1..3, kind=Ranged)`
- `Jump` → typically `JumpIntent` (can vary by design)

5.4 Not aiming (L1 not held)

- `ContextGrabOrFire` (R2) → `GrabOrClimbIntent` if opportunity exists, else fallback to `ContextInteractIntent` (optional)

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6. Intent Categories & Arbitration Policy

6.1 Intent categories

Define categories for conflict resolution:

- **Emergency**: `DodgeIntent`, `BlockIntent` (future), `JumpIntent` (sometimes)
- **Traversal**: `GrabOrClimbIntent`, `MantleIntent`
- **Combat**: `LightAttackIntent`, `HeavyAttackIntent`, `UseSkillIntent`
- **Social**: `ReasonIntent`, `BribeIntent`, `ThreatenIntent`
- **Interaction**: `LockpickIntent`, `PickpocketIntent`, `TalkIntent`, `LootIntent`

6.2 High-level arbitration rules (MVP)

1. At most **one major action** intent per tick (ActionSystem owns this)
2. Traversal intents may preempt combat if they are forced by environment (e.g., mantle)
3. Social intents are typically **lower priority** than emergency and combat, unless explicitly allowed by rules
4. If ActionSystem is locked:
    - buffer eligible intents with short TTL (e.g., 0.2–0.5s)
    - reject non-bufferable intents

6.3 Consumption model

An intent can be:

- **consumed** by a domain (no other domain processes it)
- **passed through** (ignored)
- **buffered** (ActionSystem captures it for later)

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7. Buffering and Cancel Windows

7.1 Intent buffering (ActionSystem-owned)

Common buffers:

- **Jump buffer** (platforming feel): if pressed slightly before landing, execute on landing
- **Dodge buffer**: allow dodge shortly after current animation window opens
- **Skill buffer**: queue a skill after unsheathe completes

Each buffered intent has:

- `IntentType`
- `Timestamp`
- `TTL` (time-to-live)
- Optional: `TargetId` (for social/interaction intents)

7.2 Cancel window policy (ActionSystem-owned)

Actions define cancel rules:

- `CancelableInto`: {Dodge, Jump, Skill, Reason, Interact…}
- `CancelStartTime`, `CancelEndTime`
- `HardLockTime` (no cancel)
- These are configured per action/weapon/skill.

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8. Canonical “Circle Button” Mapping (InteractionResolver)

8.1 Philosophy

“Circle” is not directly “talk” or “interact”.
Circle is **ContextInteractIntent**, which is resolved by checking opportunities.

8.2 Priority table (recommended baseline)

When `ContextInteractIntent` occurs, resolve in this order:

1. **Lockpick**: facing locked door/container and within range and have required item
2. **Stealth Takedown (Lethal)**: behind unaware target, posture eligible, lethal allowed
3. **Stealth Takedown (Non-lethal)**: behind unaware target, posture eligible
4. **Pickpocket**: near unaware NPC, posture eligible
5. **Reason (Combat Social)**: hostile target in reason range, and rule allows attempt
6. **Talk**: friendly/neutral NPC in talk range
7. **Loot/Use**: interactable object in range (chest, lever)
8. **Fallback**: no-op

8.3 Context inputs required

- `InteractionContextSnapshot` must provide best candidates:
  - `LockedDoorCandidate`
  - `NpcCandidate` + hostility + awareness + behind-angle result
  - `TalkCandidate`
  - `LootCandidate`
  - `ReasonCandidate` (hostile within shout range)

8.4 Output intents

The resolver outputs specific intents:

- `LockpickIntent(doorId)`
- `AssassinateIntent(npcId)`
- `KnockoutIntent(npcId)`
- `PickpocketIntent(npcId)`
- `ReasonIntent(npcId)`
- `TalkIntent(npcId)`
- `LootIntent(objectId)`

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9. Social Actions During Combat ("Reason")

9.1 Why "Reason" is an Action

Reasoning is:

- time-occupying
- risky (vulnerable window)
- potentially interruptible
- has cooldown and eligibility
  Therefore it belongs in ActionSystem, not in InteractionResolver.

9.2 Flow

1. `ContextInteractIntent` occurs
2. InteractionResolver returns `ReasonIntent(targetId)` if eligible and prioritized
3. ActionSystem decides:
   - start immediately (if allowed)
   - buffer (if in recover window)
   - reject (if stunned, locked, out of range)
4. ActionSystem emits `SocialAttempted` event for NPC AI/dialogue system:
   - includes player stats (int/wis/cha/luck), reputation, faction fame, recognition flags
5. NPC AI responds with an outcome:
   - pause/hesitate, demand money, retreat, attack anyway, etc.

9.3 Stats/Reputation integration

The SocialAttempt payload includes:

- player stats (int/wis/cha/luck)
- sins/virtues profile

- faction standing + fame/infamy
- recognition (e.g., “colors”)

NPC AI consumes this and decides outcome. Player systems do not hardcode NPC psychology.

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10. Stealth Integration Details

10.1 StealthTelemetry (continuous)

Produces:

- `NoiseEmission`
- `VisibilityEmission`

Noise depends on:

- posture base multiplier
- speed curve
- surface factor
- gear factor

10.2 Opportunity detection

StealthOpportunityDetector computes:

- `CanPickpocket(npcId)`
- `CanAssassinate(npcId)`
- `CanKnockout(npcId)`

Inputs:

- posture, velocity
- behind-angle check
- NPC awareness state (from world snapshot)
- distance thresholds

Outputs are consumed by InteractionResolver (for Circle mapping).

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11. Aiming / Modifier Semantics (recommended approach)

11.1 Prefer explicit modifiers over chord-named actions

Even if Unity can emit “R1+Square = SkillSlot1” as a single action, prefer:

- `SkillModifierHeld`
- `PrimaryPressed`

Reason:

- portability to other engines
- consistent resolution logic
- easier to adjust mapping rules (weapon-dependent)

11.2 Combat mode derived from AimModifier

- `AimModifierHeld` implies aiming mode candidate
- CombatMode domain can enforce additional rules:
  - cannot aim while climbing
  - cannot aim while stunned
  - aim requires bow equipped

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12. Testability Targets (what to unit test early)

1. IntentResolver mapping:
   - R1 held maps face buttons to melee skills
   - L1 held maps R2 to basic fire and face buttons to ranged skills
2. InteractionResolver priority:
   - lockpick beats talk beats loot
   - stealth takedown beats reason
3. Arbiter rules:
   - emergency intents beat social intents
4. Action buffer TTL behavior:
   - jump buffer triggers on landing
   - skill buffered behind unsheathe

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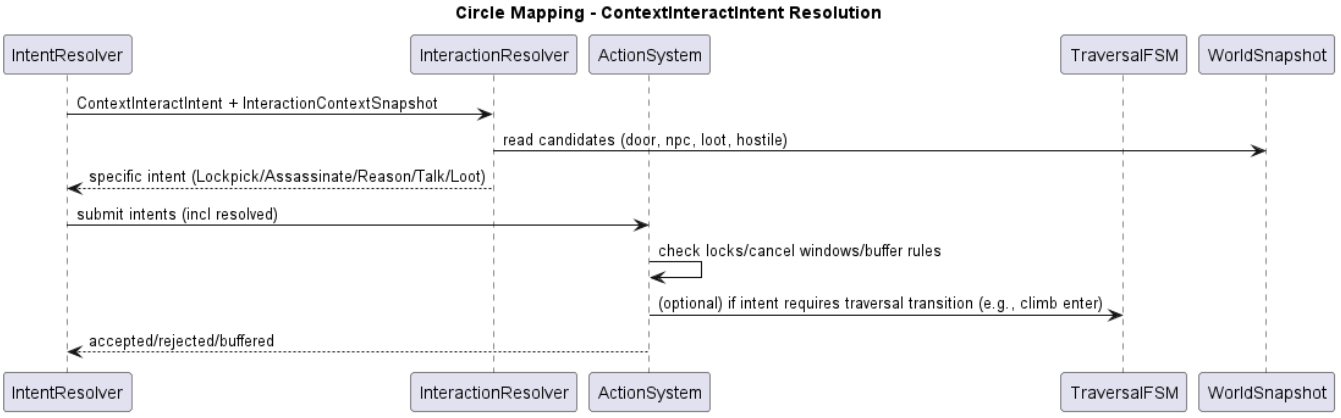
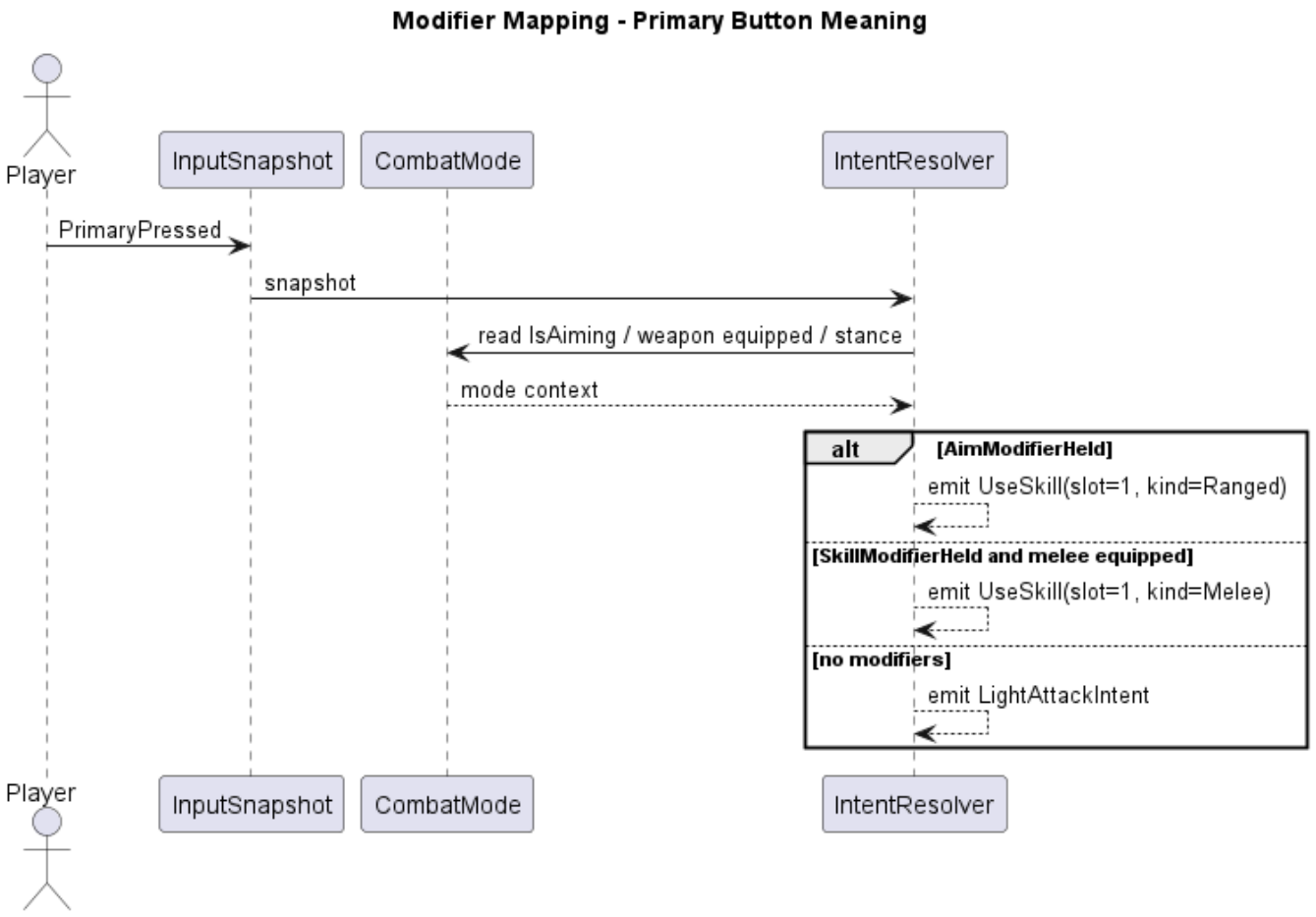
13. PlantUML Diagrams

13.1 Detailed “Circle mapping” sequence



```
@startuml
title Circle Mapping - ContextInteractIntent Resolution

participant IntentResolver as IR
participant InteractionResolver as IX
participant ActionSystem as AS
participant TraversalFSM as TR
participant WorldSnapshot as WS

IR -> IX : ContextInteractIntent + InteractionContextSnapshot
IX -> WS : read candidates (door, npc, loot, hostile)
IX --> IR : specific intent (Lockpick/Assassinate/Reason/Talk/Loot)

IR -> AS : submit intents (incl resolved)
AS -> AS : check locks/cancel windows/buffer rules
AS -> TR : (optional) if intent requires traversal transition (e.g., climb enter)
AS --> IR : accepted/rejected/buffered

@enduml
```

13.2 Modifier mapping (Dogma-inspired)




```
@startuml
title Modifier Mapping - Primary Button Meaning
```

```
actor Player
participant InputSnapshot as IN
participant CombatMode as CM
participant IntentResolver as IR

Player -> IN : PrimaryPressed
IN -> IR : snapshot
IR -> CM : read IsAiming / weapon equipped / stance
CM --> IR : mode context

alt AimModifierHeld
  IR --> IR : emit UseSkill(slot=1, kind=Ranged)
else SkillModifierHeld and melee equipped
  IR --> IR : emit UseSkill(slot=1, kind=Melee)
else no modifiers
  IR --> IR : emit LightAttackIntent
end

@enduml
```

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14. Guardrails

1. TraversalFSM must not decide combat skills.
2. ActionSystem is the only owner of action buffering and cancel windows.
3. InteractionResolver decides “which interaction,” ActionSystem decides “when/how executed.”
4. Social actions are actions, not UI states.
5. Every new interaction must be placed into the Circle priority table explicitly.